



Science PSLE 2022

Bachelor of Science general (Botswana Open University)



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BOTSWANA EXAMINATIONS COUNCIL
PRIMARY SCHOOL LEAVING EXAMINATION

SCIENCE

04

Paper 1

2022

Marks: 60

TIME: 60 Minutes

INSTRUCTIONS

1. Answer all questions on the answer sheet provided.
2. When you are told to begin, work as fast and as accurately as you can. If you cannot answer a question, do not spend too much time on it; go to the next question and come back to it later.
3. You may do rough work on the test paper or on a sheet of blank paper. **DO NOT** use the answer sheet for rough work.
4. Four possible answers are given for each question. Select the answer you think is the best and fill the oval for that answer on your answer sheet.

5. **Be sure to fill the ovals like this:**

Erase completely answers that you wish to change.

DO NOT make any stray marks on your answer sheet.

If more than one oval is filled for a question it will be marked wrong.

Be sure the oval you fill on your answer sheet is for the question you are working on in the test paper.

DO NOT mark the ovals like this:

If you do, your answers will be marked wrong.

6. Sample questions are given to help you. Read them carefully before doing the questions.

YES

17	(A)	(B)	(C)	(D)
18	(A)	(B)	(C)	(D)
19	(A)	(B)	(C)	(D)

NO

1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)

SAMPLE QUESTIONS

1. Which type of soil contains the largest particles?

A Clay
B Loam
C Sandy
D Silt

(A) (B) (C) (D)

2. Which of the following substances is **not** soluble in water?

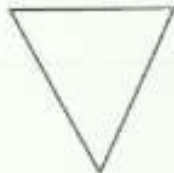
A Alcohol
B Flour
C Table salt
D Sugar

(A) (B) (C) (D)

1. Which of the following bodies of the solar system is a star?

- A Earth
- B Mars
- C Moon
- D Sun

2. The diagram below shows a weather symbol.



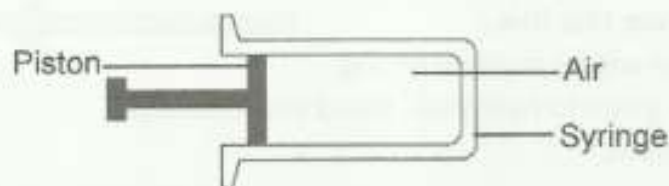
Which weather condition does the symbol represent?

- A Clear skies
- B Light drizzle
- C Strong winds
- D Thunderstorms

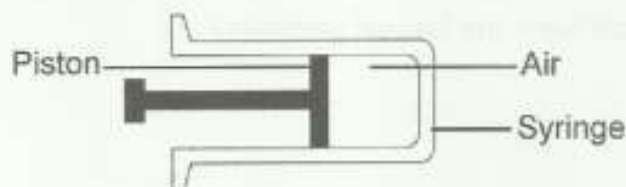
3. Which of the following is **not** a source of surface water?

- A Borehole
- B Dam
- C River
- D Lake

4. The diagram below shows a syringe containing air.



The piston can be pushed into the syringe up to a certain point as shown below.



Which property of air is being demonstrated?

- A Air supports burning.
- B Air is a mixture of gases.
- C Air has spaces between particles.
- D Air is necessary for photosynthesis.



5. Which of the following objects is made out of metal?
- A Window pane
 - B Electric wire
 - C Eraser
 - D Shoes
6. Which property of aluminium makes it suitable for making bodies of aeroplanes?
- A Having low density
 - B Having a shiny surface
 - C Being a non-magnetic material
 - D Being a good electrical conductor
7. A cup is half filled with milk and put in a bowl as shown below.



Ice is then poured into the bowl around the cup.

Which statement describes what will happen to the milk?

- A The mass of the milk will decrease.
 - B The weight of the milk will increase.
 - C The volume of the milk will decrease.
 - D The temperature of the milk will increase.
8. Which form of energy is stored on a stretched spring?
- A Kinetic energy
 - B Chemical energy
 - C Elastic potential energy
 - D Gravitational potential energy
9. Which of the following energy resources is non-renewable?
- A Coal
 - B Hydropower
 - C Solar
 - D Wind

10. Which of the following metals is a magnetic material?

- A Copper
- B Gold
- C Iron
- D Zinc

11. Which part of the eye is responsible for controlling the size of the pupil?

- A Cornea
- B Iris
- C Lens
- D Retina

12. Which of the following farm products is a body building food?

- A Beans
- B Maize
- C Sweet reeds
- D Water melon

13. What is the purpose of wearing a seat belt when driving a car?

- A To impress friends
- B To please the police
- C For increasing the speed of the car
- D For protection in case the car crashes

14. Which body system transports nutrients to the rest of the body?

- A Circulatory system
- B Digestive system
- C Nervous system
- D Skeletal system

15. Which of the following diseases is sexually transmitted?

- A Bilhazia
- B Measles
- C Herpes
- D Tuberculosis

16. Which scientist defined the concept of gravitational force?

- A Albert Einstein
- B Isaac Newton
- C Michael faraday
- D Thomas Edison

17. In which of the following fields is scientific knowledge applied the most?

- A Accounting
- B Nursing
- C Legal practice
- D Social work

Use the information below to answer question 18.

A scientist studying organisms in a forest finds that leopards feed on springboks and springboks feed on grass. The scientist also realises that the amount of grass in the forest is decreasing rapidly.

18. What is likely to happen to the number of springboks and leopards when the amount of grass decreases?

	Number of springboks	Number of leopards
A	Decrease	Decrease
B	Decrease	Increase
C	Increase	Decrease
D	Increase	Increase

19. Which term is used to describe the interaction among producers, consumers and their environment?

- A Community
- B Ecosystem
- C Habitat
- D Population

Use the information below to answer question 20.

At a kgotla meeting held to discuss environmental problems in Garona village, the community raised a concern about the waste material coming from the newly opened bottle store.

20. What advice would you give to Garona community to solve the problem they are facing?

- A Shut down the bottle store.
- B Replace the bottle store with a bar.
- C Report the business owner to the police.
- D Recycle used materials coming from the bottle store.



21. Sea water is not suitable to drink because it contains a lot of dissolved salts.

Which process can be used to get drinking water from sea water?

- A Distillation
- B Filtration
- C Freezing
- D Boiling

22. In which diagram below will the salt completely dissolve first when the flasks are shaken?

A



250 ml of
water at 50°C

4 g of coarse salt

B



250 ml of
water at 10°C

4 g of coarse salt

C



250 ml of
water at 50°C

4 g of powdered
salt

D



250 ml of
water at 10°C

4 g of powdered
salt

23. A water supply company claims that the water they provide to homes is soft.

Which of the following observations proves that the company is telling the truth?

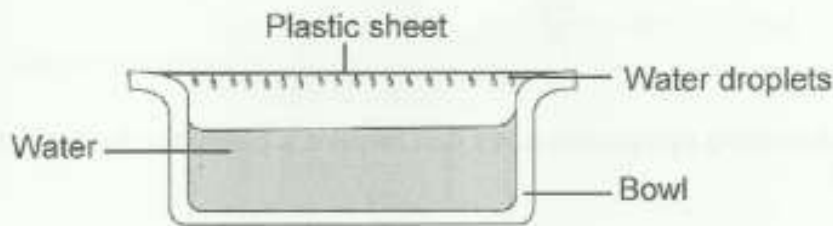
- A The water causes fur deposits to form in kettles.
- B The water causes blockage of drip irrigation hoses.
- C The water leads to narrowing of pipes fitted in houses.
- D The water requires a small amount of soap to form lather.

24. Hot tea was poured into four cups made of different materials. The materials used were plastic, steel, copper and aluminium.

In which cup will the tea remain hot for the longest time?

- A Aluminium cup
- B Copper cup
- C Plastic cup
- D Steel cup

25. A bowl is filled with water and covered with a plastic sheet. When the bowl is put under direct sunlight, water droplets form on the plastic sheet as shown in the diagram.



Which processes led to the movement of water from the bowl to the plastic sheet?

- A Boiling followed by freezing
B Freezing followed by melting
C Melting followed by condensation
D Evaporation followed by condensation
26. Which of the following processes results in a decrease of temperature?
- A Boiling
B Condensation
C Evaporation
D Melting
27. In which part of the female reproductive system does the zygote form?
- A Cervix
B Oviduct
C Uterus
D Vagina
28. What is the main energy change that takes place when a person shouts?
- A Strain energy → kinetic energy
B Sound energy → heat energy
C Chemical energy → kinetic energy
D Chemical energy → sound energy
29. Why are hot water storage tanks painted silver?
- A To increase heat loss by radiation
B To increase heat gain by radiation
C To decrease heat loss by radiation
D To decrease heat gain by radiation
30. Why do car manufacturers fit mirrors on the cars?
- A To capture images of objects on the road
B To magnify letters when drivers are driving
C For drivers to view behind the cars when driving
D For drivers to view distant objects in front of the car

31. Sound travels fastest through which of the following substances?

- A Air
- B Oil
- C Soil
- D Water

32. Which of the following symptoms does **not** require a patient to be tested for tuberculosis?

- A High fever
- B Losing weight
- C Coughing blood
- D Frequent sneezing

33. A red litmus paper and a blue litmus paper are both put in a bowl containing vinegar.

Which row in the table below shows how each litmus paper will be affected by the vinegar?

	Blue litmus paper	Red litmus paper
A	Remains blue	Remains red
B	Remains blue	Turns blue
C	Turns red	Remains red
D	Turns red	Turns blue

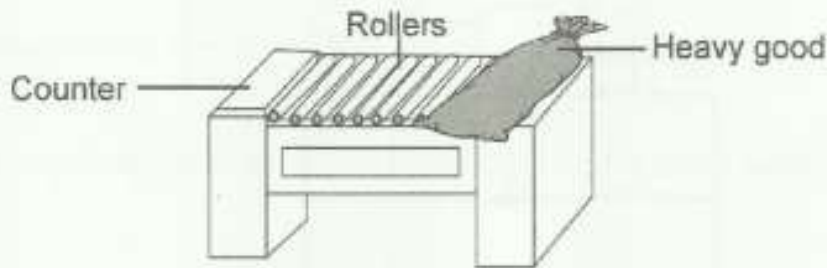
34. Which organ is responsible for the absorption of nutrients from digested food into the blood stream?

- A Liver
- B Pancreas
- C Large intestine
- D Small intestines

35. Which of the following diseases are children immunised against?

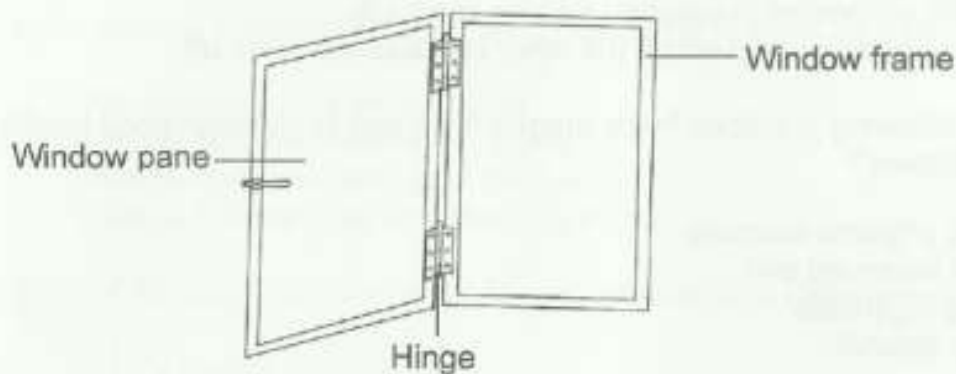
- A AIDS
- B Gonorrhoea
- C Measles
- D Syphilis

36. In wholesalers, heavy goods are moved over rollers fitted to counters at payment points as shown in the diagram below.



What is the use of the fitted rollers on the counter?

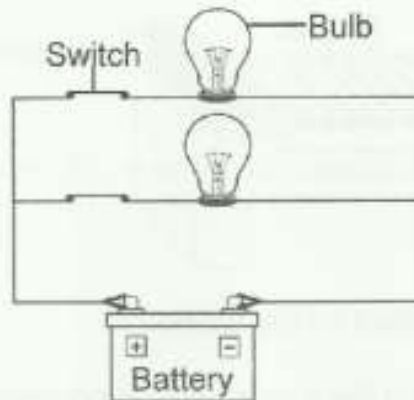
- A To reduce friction between the goods and the counter
 - B To reduce the speed of the goods across the counter
 - C To increase the stability of the goods
 - D To increase the weight of the goods
37. The diagram below shows an open window.



To which class of simple machines does the hinge belong?

- A Gears
 - B Levers
 - C Pulleys
 - D Wheels
38. Which of the following substances is soluble in water?
- A Maize meal
 - B Iron filings
 - C Sand
 - D Coffee

39. The diagram below shows a parallel connection of two bulbs and two switches to a battery to make a lighting circuit.



What is the advantage of connecting the lighting circuit in parallel instead of in series?

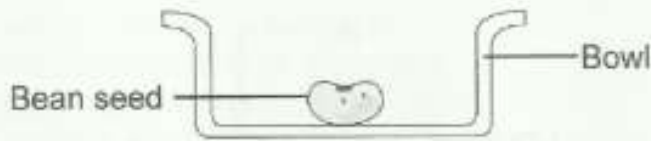
- A The bulbs connected in parallel last longer.
 - B When one bulb blows the other one remains lit.
 - C The bulbs connected in parallel use less electricity.
 - D When one bulb is switched off the other one also switches off.
40. Which of the following practices helps pregnant women to maintain good health of their unborn babies?
- A Avoiding physical exercise
 - B Eating a balanced diet
 - C Smoking cigarettes
 - D Drinking alcohol

Use the information below to answer question 41.

Everyday when Betty comes back from school she sits in front of the television, eating and playing video games.

41. Which of the following problems is Betty likely to develop?
- A Overweight
 - B Loneliness
 - C Tiredness
 - D Backache
42. Which term describes organisms that breakdown dead plants into nutrients in the soil?
- A Producers
 - B Consumers
 - C Scavengers
 - D Decomposers

43. The diagram below shows a bowl containing a bean seed.



Which substance has to be added to the bowl for the bean seed to germinate?

- A Oil
- B Soil
- C Water
- D Carbon dioxide

44. What are the characteristics of fruits that are dispersed by wind?

- A They have many hooks.
- B They are fresh and juicy.
- C They are colourful and bright.
- D They have a parachute-like structure.

45. Which change is caused by the rotation of the earth about its axis?

- A Change of months within a year
- B Change between seasons in a year
- C Change from one decade to the next
- D Change between day-time and night-time

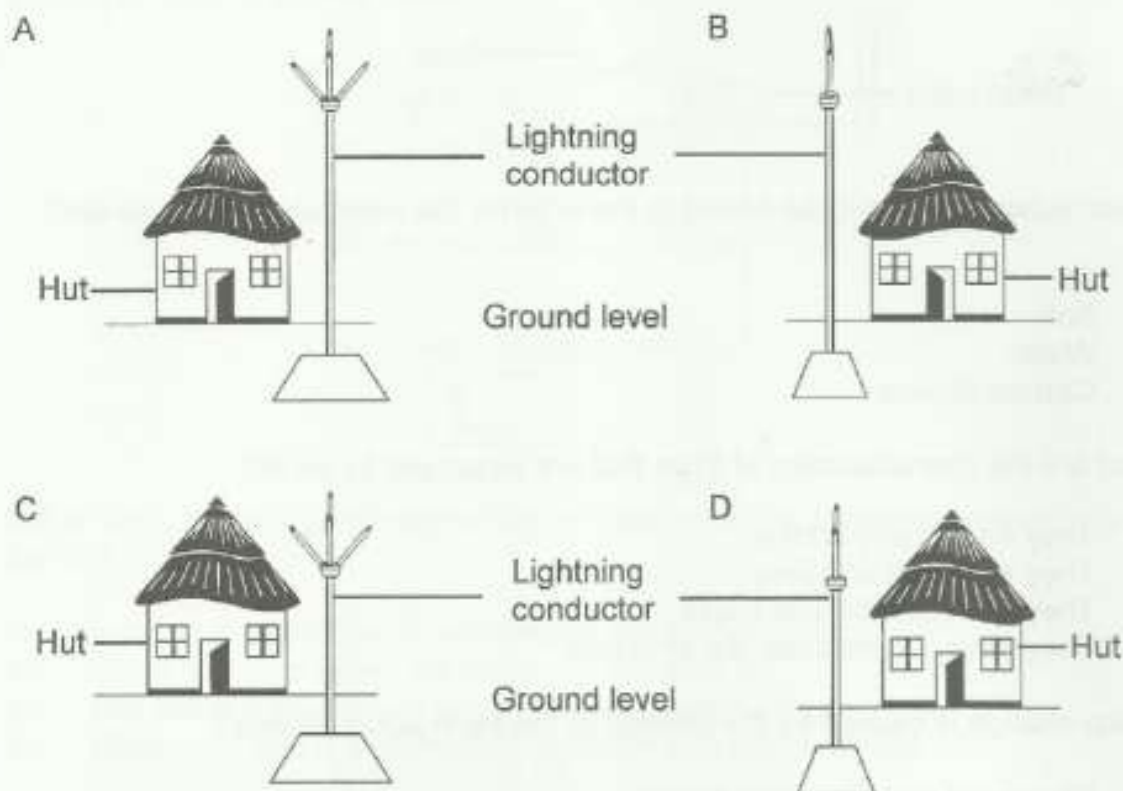
46. Which of the properties below are for plants found in a desert environment?

- A Thorny leaves
- B Fleshy stems
- C Short roots
- D Big leaves

47. Which component works by melting when too much current flows through the circuit?

- A Fuse
- B Switch
- C Live wire
- D Earth wire

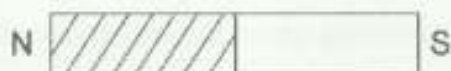
48. In which of the following situations would the lightning conductor best protect the hut?



49. Which of the following appliances uses a motor to perform its function?

- A Iron
- B Drill
- C Lamp
- D Stove

50. The diagram below shows a bar magnet with its north pole (N) and south pole (S) labelled.



The magnet is broken in the middle and the two pieces separated as shown below.



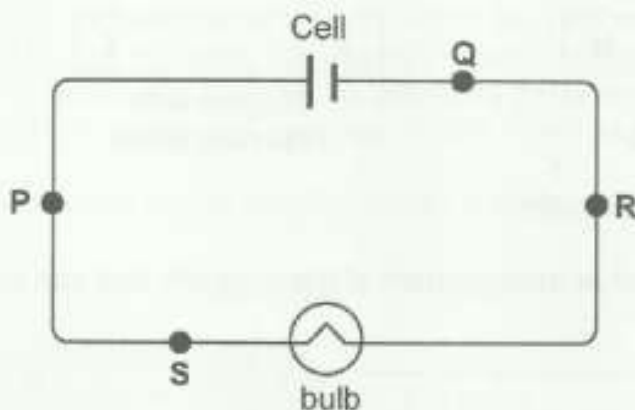
What are the resulting poles on the pieces at end X and end Y?

	Pole at end X	Pole at end Y
A	North	North
B	North	South
C	South	North
D	South	South

51. Which of the following activities is safe to do in a science laboratory?

- A Tasting unknown substances
- B Inhaling vapours from chemicals
- C Handling insulated electrical cables
- D Touching chemicals with bare hands

52. The diagram below shows a circuit consisting of a cell and bulb. Four points P, Q, R and S are marked along the circuit.



Which two points can be joined with a conducting wire without causing a short circuit?

- A P and Q
- B Q and R
- C R and S
- D Q and S

Use the information below to answer question 53.

Tebza is the first to arrive at an accident scene. She immediately decides to assist the accident victims before the arrival of the medical crew.

53. What should Tebza do before assisting the victims?

- A Wear overalls
- B Put on a helmet
- C Put on some gumboots
- D Wear some surgical gloves

54. What is the advantage of thermal power stations over hydroelectric power stations?

- A Thermal power stations do not pollute the air.
- B Thermal power stations are a renewable energy resource.
- C Thermal power stations can be built anywhere in the country.
- D Thermal power stations are cheaper to operate over a long period of time.

55. In which of the following cases does a patient have a disease?

- A A patient is long sighted.
- B A patient's arm is broken.
- C A patient ate a poisonous plant.
- D A patient has harmful bacteria in the blood.

56. The diagrams below show a magnet whose poles are known, placed next to another magnet whose poles are not known.



Magnet with
known poles



Magnet with
unknown poles

The two magnets attract each other.

Which diagram shows another arrangement of the magnets that can make them attract each other?

A	<table border="1" style="display: inline-table; margin-right: 20px;"> <tr> <td style="padding: 5px;">S</td><td style="padding: 5px;">N</td></tr> </table> <table border="1" style="display: inline-table;"> <tr> <td style="padding: 5px;">Y</td><td style="padding: 5px;">X</td></tr> </table>	S	N	Y	X
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N	S				
X	Y				

57. A girl accidentally touches a hot stove and suddenly pulls her hand away.

Which part of the nervous system is **not** involved in the girl's action?

- A Brain
- B Spinal cord
- C Motor neurone
- D Sensory neurone



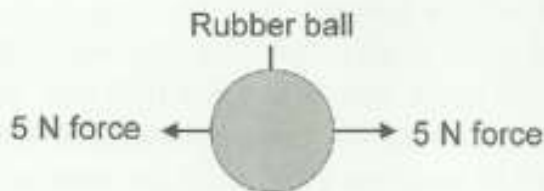
58. Why are black animals **more** likely to be hit by vehicles at night than white animals?

- A White colour reflects more light.
- B Black colour absorbs more heat.
- C Black animals have poor eye sight.
- D White animals have better hearing.

59. How does having sexually transmitted diseases (STDs) increase the risk of contracting HIV?

- A All STDs are caused by viruses which are similar to HIV.
- B All STDs are incurable just like HIV which causes incurable AIDs.
- C Most STDs cause fever which weakens the body for HIV infection.
- D Most STDs cause sores on sex organs which exposes them for HIV infection.

60. The diagram below shows two 5 N forces pulling on a rubber ball.



Which property of the rubber ball will be changed by the forces?

- A Mass
- B Speed
- C Shape
- D Direction